

MAR DEL PLATA, Argentina

WMO: 876920

Lat: 37.9322S

Lon: 57.5814W

Elev: 69

StdP: 14.66

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.0	32.2	23.5	17.8	38.3	26.6	20.6	38.5	26.0	50.4	23.2	49.7	4.5	320	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.8	88.3	70.2	84.8	68.9	81.4	68.0	73.7	82.3	72.1	79.7	70.7	77.6	15.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	116.3	76.8	69.8	109.9	75.2	68.3	104.1	73.7	37.4	82.1	35.9	79.5	34.7	77.7	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	21.2	19.0	DB	25.9	96.2	2.4	3.0	24.2	98.3	22.7	100.1	21.4	101.8	19.6	103.9	(4)
(5)			WB	25.0	77.5	2.6	2.5	23.1	79.3	21.6	80.8	20.2	82.1	18.4	83.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	68.5	67.4	64.3	58.4	53.0	47.7	46.3	48.7	50.7	55.7	60.5	65.8
	DBStd	9.58	5.88	5.88	5.98	5.64	5.53	4.98	5.23	5.54	5.33	5.74	5.95	6.59
	HDD50	436	0	0	0	4	30	102	140	91	54	14	1	0
	HDD65	3299	33	39	88	210	372	518	580	506	428	294	160	71
	CDD50	3067	574	487	445	255	124	34	25	50	76	190	316	491
	CDD65	457	142	107	67	11	2	0	0	1	0	5	25	97
	CDH74	3364	1030	680	367	66	10	0	0	10	7	70	258	866
	CDH80	1038	344	207	87	8	0	0	0	1	1	9	59	322
Wind	WSAvg	9.1	10.0	9.4	8.7	8.1	7.4	8.4	8.3	9.0	9.4	9.8	10.0	10.4
Precipitation	PrecAvg	35.5	3.8	3.1	3.9	3.3	2.6	2.4	2.2	2.6	2.3	3.3	3.0	3.7
	PrecMax	55.3	8.1	11.9	10.5	14.6	9.4	6.9	6.8	7.7	6.2	10.0	7.4	13.0
	PrecMin	22.0	0.8	0.4	0.3	0.0	0.2	0.1	0.0	0.1	0.2	0.2	0.7	0.2
	PrecStd	7.5	2.0	1.9	2.2	3.1	1.7	1.6	1.5	2.0	1.4	2.0	1.6	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.2	91.2	87.7	80.2	75.1	66.2	66.6	73.6	73.5	80.7	86.1	93.3
		MCWB	72.2	72.5	70.5	66.9	64.0	58.2	58.4	61.4	60.4	65.3	66.5	71.4
	2%	DB	87.9	86.0	82.1	75.3	68.7	62.4	61.5	66.1	68.2	75.2	80.8	87.8
		MCWB	70.8	70.6	69.2	64.8	60.7	55.8	55.0	57.0	57.1	62.2	65.4	69.0
	5%	DB	83.7	81.9	77.8	71.7	65.3	59.3	58.1	62.0	64.4	70.8	76.5	83.0
		MCWB	69.2	69.3	67.4	63.0	58.5	54.0	52.3	53.6	55.2	60.1	63.8	67.4
	10%	DB	79.6	77.6	74.4	68.4	62.5	57.2	55.3	58.6	60.8	66.4	71.9	78.6
		MCWB	68.0	68.5	66.7	62.3	57.6	52.4	50.8	51.8	53.5	57.9	62.0	66.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	75.9	74.1	70.5	66.1	62.8	61.9	64.2	63.2	68.5	70.0	74.7
		MCDB	86.1	84.3	80.7	74.9	71.2	64.3	65.0	69.5	69.3	77.0	79.5	86.7
	2%	WB	73.3	73.6	71.9	67.8	63.0	58.6	56.8	58.8	59.3	64.3	67.5	71.9
		MCDB	82.6	80.8	77.9	71.2	66.6	60.4	59.5	63.6	65.1	72.3	76.3	82.2
	5%	WB	71.7	71.8	69.9	65.4	60.8	55.7	53.8	55.5	57.0	61.8	65.5	69.8
		MCDB	79.7	77.9	75.1	69.4	63.5	58.0	56.3	59.7	62.1	68.1	73.0	78.4
	10%	WB	70.2	70.2	68.1	63.3	58.7	53.3	51.8	53.1	54.8	59.4	63.7	67.7
		MCDB	77.4	75.8	72.8	67.6	61.6	56.2	54.5	57.0	59.0	64.8	70.3	75.4
Mean Daily Temperature Range	5% DB	MDBR	19.8	18.2	17.9	18.2	17.0	16.7	16.3	17.2	17.4	17.4	19.3	20.6
		MCDBR	27.4	25.3	23.5	23.5	21.5	19.8	20.5	23.7	24.4	26.2	27.4	29.7
		MCWBR	13.4	12.6	12.5	13.9	13.8	13.9	14.1	15.0	14.9	15.1	14.5	14.0
	5% WB	MCDBR	22.0	19.6	18.2	18.2	16.6	14.6	15.6	18.8	20.1	21.4	22.1	24.0
		MCWBR	12.2	11.4	11.2	12.3	12.2	12.3	12.2	13.3	13.7	13.8	13.7	13.2
Clear-Sky Solar Irradiance	taub	0.402	0.384	0.363	0.356	0.346	0.339	0.347	0.397	0.419	0.391	0.376	0.386	
	taud	2.362	2.429	2.481	2.451	2.445	2.451	2.402	2.226	2.176	2.321	2.394	2.384	
	Ebn at Noon	295	293	286	267	249	240	245	246	262	287	301	302	
	Edh at Noon	41	37	32	29	25	23	26	36	43	41	40	41	
All-Sky Solar Radiation	RadAvg	2286	1976	1558	1119	758	608	656	894	1294	1701	2106	2390	
	RadStd	95	132	98	115	67	56	56	96	98	164	108	121	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.22	N/A	N/A	N/A	N/A	N/A	+52
(47) Regional (2 neighbors)	N/A	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	N/A	+59

Nomenclature: See separate page